

5-1

*n*th Roots, Radicals,
and Rational
Exponents

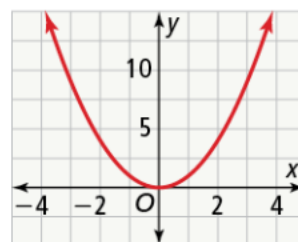
I CAN... relate roots and rational exponents and use them to simplify expressions and solve equations.

VOCABULARY

- n*th root

EXPLORE & REASON

The graph shows $y = x^2$.



- A.** Find *all* possible values of x or y so that the point is on the graph.
- | | |
|--------------------------------------|---------------------------------------|
| (a) $(2, \underline{\hspace{1cm}})$ | (b) $(3, \underline{\hspace{1cm}})$ |
| (c) $(-3, \underline{\hspace{1cm}})$ | (d) $(5, \underline{\hspace{1cm}})$ |
| (e) $(\underline{\hspace{1cm}}, 4)$ | (f) $(\underline{\hspace{1cm}}, -16)$ |
| (g) $(\underline{\hspace{1cm}}, 7)$ | (h) $(\underline{\hspace{1cm}}, 5)$ |
- B. Communicate Precisely** Write a precise set of instructions that show how to find an approximate value of $\sqrt{13}$ using the graph.
- C.** Draw a graph of $y = x^3$. Use the graph to approximate each value.
- | | |
|------------------------------|-----------------------------|
| (a) $\sqrt[3]{5}$ | (b) $\sqrt[3]{-5}$ |
| (c) $\sqrt[3]{8}$ | (d) A solution to $x^3 = 5$ |
| (e) A solution to $x^3 = -5$ | (f) A solution to $x^3 = 8$ |

ESSENTIAL QUESTION

How are exponents and radicals used to represent roots of real numbers?

EXAMPLE 1 Find all Real *n*th Roots

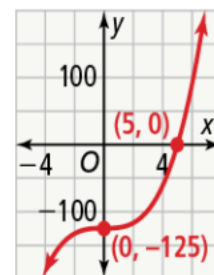
- A.** What are all the real cube roots of 125?

An ***n*th root** of a number c is x , such that $x^n = c$. An *n*th root can be represented using a radical symbol by $x = \sqrt[n]{c}$.

Since $125 = 5^3$, $\sqrt[3]{125} = 5$ is a real cube root.

To determine if there are other real cube roots of 125, consider the function $f(x) = x^3 - 125$. The zeros of f are the cube roots of 125.

The graph of f has only one x -intercept. Therefore, 5 is the only real cube root of 125.

**MAKE SENSE AND PERSEVERE**

Recall that third degree polynomial functions have three solutions. Here, one solution is real and the other two are complex. How could you find the complex solutions?

CONTINUED ON THE NEXT PAGE

CONSTRUCT ARGUMENTS

Consider the function $f(x) = x^4 - 16$. How can you use the graph of f to conclude that 16 has two real fourth roots? © MP.3

EXAMPLE 1 CONTINUED**B. What are all the real fourth roots of 16?**

Since $16 = 2^4$, one fourth root is the principal root, $\sqrt[4]{16} = 2$.

To determine all the real fourth roots of 16, solve $x^4 = 16$.

$$x^4 = 16$$

$$x^4 - 16 = 0$$

$$(x^2 - 4)(x^2 + 4) = 0$$

$$(x - 2)(x + 2)(x^2 + 4) = 0$$

The zeros of the first two factors are 2 and -2 . The zeros of the quadratic factor $x^2 + 4$ are imaginary. Therefore, the real fourth roots of 16 are 2 and -2 .

Try It! 1. Find the specified roots of each number.

a. real fourth roots of 81

b. real cube roots of 64

CONCEPTUAL UNDERSTANDING**STUDY TIP**

It is important to recognize multiple representations of the same value so you can select the form that is most efficient for a given situation.

EXAMPLE 2 Understand Rational Exponents**A. What is the meaning of the exponent in the expression $16^{\frac{1}{4}}$?**

Extend the properties of *integer* exponents to *rational* exponents.

$$16^{\frac{1}{4}} = (2^4)^{\frac{1}{4}}$$

$$= 2^1$$

$$= \sqrt[4]{16}$$

Use the Power of a Power Property.

Since $2 = \sqrt[4]{16}$, $16 = 2^4$.

Therefore, $16^{\frac{1}{4}}$ is another way to represent $\sqrt[4]{16}$, or 2.

In general, for positive integer n , $x^{\frac{1}{n}}$ can be expressed as the radical $\sqrt[n]{x}$.

$$x^{\frac{1}{n}} = \sqrt[n]{x}$$

Both the exponent $\frac{1}{n}$ and the radical symbol $\sqrt[n]{x}$ indicate the principal n th root.

B. Write $27^{\frac{2}{3}}$ as a radical.

$$27^{\frac{2}{3}} = (27^{\frac{1}{3}})^2$$

$$= (\sqrt[3]{27})^2$$

$$= 3^2$$

$$= 9$$

Use the Power of a Power Property.

This means that you can rewrite $27^{\frac{2}{3}}$ as $(\sqrt[3]{27})^2$. Since 27 is a perfect cube, you can evaluate the radical expression mentally by taking the cube root first and then squaring.

$$27^{\frac{2}{3}} = (\sqrt[3]{27})^2 = 3^2 = 9$$

Try It! 2. Explain what each fractional exponent means, then evaluate.

a. $25^{\frac{1}{2}}$

b. $32^{\frac{2}{5}}$

CONCEPT Interpreting Rational Exponents

In general, if the n th root of c is a real number, $\sqrt[n]{c} = c^{\frac{1}{n}}$.

Furthermore, if m is an integer, then

$$c^{\frac{m}{n}} = (c^{\frac{1}{n}})^m = (\sqrt[n]{c})^m \text{ and } \sqrt[n]{c^m} = (c^m)^{\frac{1}{n}} = c^{\frac{m}{n}}.$$

APPLICATION

USE APPROPRIATE TOOLS

Many expressions with integer roots can be evaluated using mental math. However, when the root is not an integer, it is helpful to use the power function of a calculator to compute rational roots. © MP5

EXAMPLE 3 Evaluate Expressions With Rational Exponents

A. Evaluate the expressions $32^{\frac{3}{5}}$, $27^{-\frac{2}{3}}$, and $50^{\frac{3}{4}}$.

$$\begin{aligned} 32^{\frac{3}{5}} &= (32^{\frac{1}{5}})^3 \\ &= 2^3 \\ &= 8 \end{aligned}$$

$$\begin{aligned} 27^{-\frac{2}{3}} &= (27^{\frac{1}{3}})^{-2} \\ &= (3)^{-2} \\ &= \frac{1}{9} \end{aligned}$$

Since 50 does not have a perfect 4th root, use a calculator to approximate:

$$50^{\frac{3}{4}} \approx 18.80$$

B. The Fujita scale rating, F , of a tornado is represented by

$F = \sqrt[3]{\left(\frac{W}{14.1}\right)^2} - 2$, where W is the estimated wind speed of the tornado in miles per hour. What is the Fujita scale rating for this tornado?



Wind speed estimated at 100 mph

$$\begin{aligned} F &= \sqrt[3]{\left(\frac{100}{14.1}\right)^2} - 2 \\ &= \left(\frac{100}{14.1}\right)^{\frac{2}{3}} - 2 \\ &\approx (7.09)^{\frac{2}{3}} - 2 \\ &\approx 3.69 - 2 \\ &\approx 1.69 \end{aligned}$$

A tornado with estimated wind speeds of 100 mph is classified as F1 according to the Fujita scale.

Try It! 3. What is the value of each expression? Round to the nearest hundredth if necessary.

a. $-(16^{\frac{3}{4}})$

b. $\sqrt[5]{3.5^4}$

EXAMPLE 4 Simplify n th Roots

Simplify each expression. Assume all variables are positive.

A. $\sqrt[5]{32m^{15}}$

$$\sqrt[5]{2^5 m^{15}} = \sqrt[5]{(2m^3)^5}$$

$$= 2m^3$$

So, $\sqrt[5]{32m^{15}} = 2m^3$.

Write the radicand as an expression to the 5th power.

B. $\sqrt[4]{x^{20}y^8}$

$$\sqrt[4]{x^{20}y^8} = (x^{20}y^8)^{\frac{1}{4}}$$

$$= x^{\frac{20}{4}}y^{\frac{8}{4}}$$

$$= x^5y^2$$

So, $\sqrt[4]{x^{20}y^8} = x^5y^2$.

Raise the radicand to the one-fourth power.

USE STRUCTURE

You can also work this problem using the method in Part A. The exponents 20 and 8 are both multiples of 4. So you can rewrite

this expression as $\sqrt[4]{x^{20}y^8} = \sqrt[4]{(x^5y^2)^4}$ which equals x^5y^2 .**Try It!** 4. Simplify each expression.

a. $\sqrt[3]{-8a^3b^9}$

b. $\sqrt[4]{256x^{12}y^{24}}$

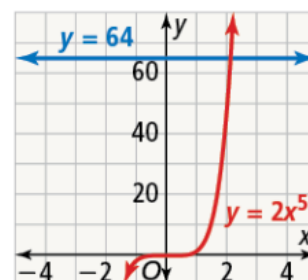
EXAMPLE 5 Use n th Roots to Solve EquationsSolve the equation $2x^5 = 64$.

A. $2x^5 = 64$

$x^5 = 32$

$(x^5)^{\frac{1}{5}} = 32^{\frac{1}{5}}$

$x = 2$

Raise each side of the equation to the $\frac{1}{5}$ power.

The graph shows one point of intersection, so there is only one real solution.

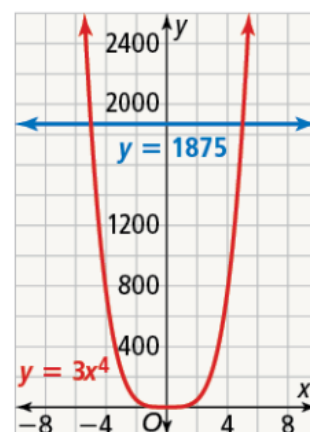
The solution to the equation is $x = 2$.

B. $3x^4 = 1875$

$x^4 = 625$

$(x^4)^{\frac{1}{4}} = \pm(625)^{\frac{1}{4}}$

$x = \pm 5$

A negative integer raised to an even power is positive, so -5 is also a real solution.

The graphs show 2 points of intersection, so there are no other real solutions.

The solutions are $x = \pm 5$.**GENERALIZE**

Raise both sides of the equation to a power so that the exponent of the variable becomes 1.

Using the Power of a Power

Property, $(x^n)^{\frac{1}{n}} = x^1$.**Try It!** 5. a. Solve the equation $5x^3 = 320$.

b. Solve the equation $2p^4 = 162$.

CONCEPT Solving an Equation in the Form $x^n = c$

To solve an equation in the form $x^n = c$, find the n th root of both sides by raising each expression to the $\frac{1}{n}$ power.

$$(x^n)^{\frac{1}{n}} = (c)^{\frac{1}{n}}$$

APPLICATION

EXAMPLE 6 Use n th Roots to Solve Problems

A garden water fountain is made of two cubes. The larger container has an edge length 2 cm longer than the edge length of a second cube. When the larger cube empties into the smaller cube, how much water will spill over the edge of the smaller cube into the base?



The volume of the larger cube is 729 cm^3 .

MODEL WITH MATHEMATICS

Use the information in the problem to write an equation that models the situation. © MP4

$$(x + 2)^3 = 729$$

$$[(x + 2)^3]^{\frac{1}{3}} = (729)^{\frac{1}{3}}$$

$$x + 2 = 9$$

$$x = 7$$

Raise both sides to the $\frac{1}{3}$ power and solve.

The volume of the smaller cube is 7^3 , or 343 cm^3 .

The volume of the larger cube is 729 cm^3 .

The amount of water that spills is $729 - 343 = 386$.

When the larger cube empties into the smaller cube, 386 cm^3 of water will spill out into the base of the fountain.

Try It! 6. One cube has an edge length 3 cm shorter than the edge length of a second cube. The volume of the smaller cube is 200 cm^3 . What is the volume of the larger cube?

CONCEPT SUMMARY n th Roots, Radicals, and Rational Exponents

Relating Radical and Exponential Forms

WORDS The index of a radical is equivalent to the denominator of a fractional exponent. The exponent of the radicand is equivalent to the numerator of a fractional exponent.

NUMBERS Radical Form $\sqrt[5]{32^4} = (32^4)^{\frac{1}{5}} = 32^{\frac{4}{5}}$

Exponential Form $729^{\frac{5}{6}} = (729^{\frac{1}{6}})^5 = (\sqrt[6]{729})^5$

ALGEBRA Radical Form $\sqrt[n]{c^m} = (c^m)^{\frac{1}{n}} = c^{\frac{m}{n}}$

Exponential Form $c^{\frac{m}{n}} = (c^{\frac{1}{n}})^m = \sqrt[n]{c^m}$

Solving an Equation in the Form $x^n = c$

To solve an equation in the form $x^n = c$, find the n th root of both sides of the equation by raising each expression to the $\frac{1}{n}$ power.

$$x^3 = 1,728$$

$$(x^3)^{\frac{1}{3}} = (1,728)^{\frac{1}{3}}$$

$$x = 12$$

$$x^n = c$$

$$(x^n)^{\frac{1}{n}} = (c)^{\frac{1}{n}}$$

$$x = c^{\frac{1}{n}}$$

Do You UNDERSTAND?

- ESSENTIAL QUESTION** How are exponents and radicals used to represent roots of real numbers?
- Error Analysis** Kaitlyn said $\sqrt[3]{10} = 10^3$. Explain Kaitlyn's error.
- Vocabulary** In the radical expression $\sqrt[5]{125}$, what is the index? What is the radicand?
- Use Structure** Why is $75^{\frac{3}{5}}$ equal to $(75^{\frac{1}{5}})^3$?
- Construct Arguments** Amani said that $(x^8)^{\frac{1}{4}} = \frac{x^8}{x^4} = x^4$. Is Amani correct? Explain.
- Make Sense and Persevere** Is it possible for a rational exponent to be an improper fraction? Explain how $27^{\frac{4}{3}}$ is evaluated or why it cannot be evaluated.

Do You KNOW HOW?

Write each expression in radical form.

7. $a^{\frac{1}{5}}$

8. $7^{\frac{2}{3}}$

Write each expression in exponential form.

9. $\sqrt[3]{b}$

10. $\sqrt[4]{p^7}$

- How many real third roots does 1,728 have?
- How many real sixth roots does 15,625 have?
- Solve the equation $4x^3 = 324$.
- Solve the equation $2x^4 = 2,500$.

Simplify each expression. Assume all variables are positive.

15. $\sqrt[3]{27x^{12}y^6}$

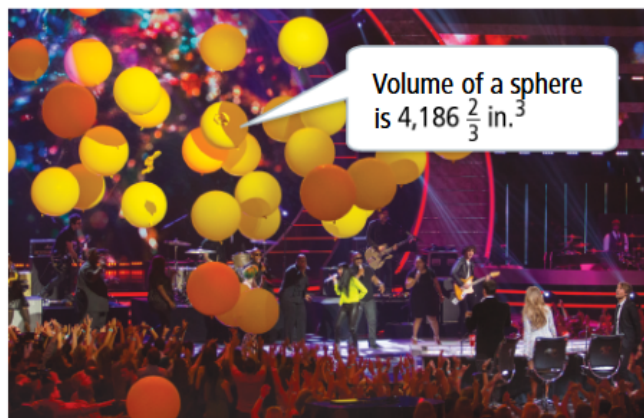
16. $\sqrt[5]{-32x^5y^{30}}$

- A snow globe is packaged in a cubic container that has volume 64 in.³ A large shipping container is also a cube, and its edge length is 8 inches longer than the edge length of the snow globe container. How many snow globes can fit into the larger shipping container?



UNDERSTAND

18. **Construct Arguments** Justice found that the fifth root of $243x^{15}y^5$ is $3x^3y$. Is Justice correct? Explain your reasoning. © MP.3
19. **Make Sense and Persevere** For a show, spheres were bounced and passed. Explain how to find the radius r of one of the inflated spheres. Use technology to compute your answer. © MP.1



20. **Error Analysis** Describe and correct the error a student made in writing this exponential expression in radical form. © MP.3

$$\begin{aligned} x^{\frac{4}{3}} &= (x^4)^{\frac{1}{3}} \\ (x^4)^{\frac{1}{3}} &= \sqrt[4]{x^3} \end{aligned}$$

X

21. **Construct Arguments** Determine whether $\sqrt[3]{x^2}$ is equal to $(\sqrt[3]{x})^2$. Explain your reasoning. © MP.3
22. **Use Structure** How many third roots does -512 have? Explain your reasoning. © MP.7
23. **Higher Order Thinking** The annual interest formula below calculates the final balance of an account, F , given a starting balance, S , and an interest rate, r , after 10 years.
- $$F = S(1 + r)^{10}$$
- When solving for r , why can the negative root be ignored?
24. **Mathematical Connections** The lengths of the two legs of a right triangle are 4 and 8. What is the length of the hypotenuse, in simplest radical form?

PRACTICE

Find the specified roots of each number.

SEE EXAMPLE 1

25. the real fourth roots of 81
26. the real third roots of 343
27. the real fifth roots of 1,024
28. the real square roots of 25

Rewrite each expression using a fractional exponent. SEE EXAMPLE 2

29. $\sqrt[4]{16^2}$ 30. $\sqrt[6]{729}$
31. $\sqrt[7]{x^2}$ 32. $\sqrt[4]{ab}$

What is the value of each expression? Round to the nearest hundredth if necessary. SEE EXAMPLE 3

33. $\sqrt[4]{25^2}$ 34. $-\sqrt[3]{125^5}$

Simplify each expression. Assume all variables are positive. SEE EXAMPLE 4

35. $\sqrt[3]{8y^9}$ 36. $\sqrt[4]{q^{12}z^4}$
37. $\sqrt[6]{729a^{24}b^{18}}$ 38. $\sqrt[8]{v^8g^{40}}$

Solve each equation. SEE EXAMPLE 5

39. $1,125 = 9x^3$ 40. $6,480 = 5w^4$
41. $270 = 10q^3$ 42. $256 = 4h^6$

43. A small cube holds chocolate candies. Its side length is 1.5 in. smaller than a second, larger cube of chocolate candies. What is the volume of the larger cube? SEE EXAMPLE 6

Volume is 85 in.³



PRACTICE & PROBLEM SOLVING

APPLY

44. **Model With Mathematics** A water-walking ball has a set volume. What is the radius, r , of the ball?



45. **Make Sense and Persevere** Ahmed received a box of gifts. The box is a rectangular prism with the same height and width, and the length is twice the width. The volume of the box is $3,456 \text{ in.}^3$. What is the height of the box?



46. **Make Sense and Persevere** Amelia's bank account earns interest annually. The equation shows her starting balance of \$200 and her balance at the end of four years, \$220.82. At what rate, r , did Amelia earn interest?

$$220.82 = 200(1 + r)^4$$

47. **Model With Mathematics** One measure of a patient's body surface area is found using the expression $\sqrt{\frac{H \cdot W}{3,600}}$. Write this with a fractional exponent.

ASSESSMENT PRACTICE

48. Determine if each expression is another way to write $b^{\frac{3}{4}}$. Select Yes or No.

	Yes	No
a. $\sqrt[4]{b^3}$	☐	☐
b. $(b^3)^{\frac{1}{4}}$	☐	☐
c. $b^{\frac{4}{3}}$	☐	☐
d. $\sqrt[3]{b^4}$	☐	☐
e. $\frac{b^3}{b^4}$	☐	☐

49. **SAT/ACT** Which of the following is equivalent to $\sqrt[6]{4,096x^{18}y^{30}}$, where $x > 0$ and $y > 0$?

- Ⓐ $682.7x^{15}y^{24}$
- Ⓑ $4x^{1.6}y^{1.8}$
- Ⓒ $4,096x^3y^5$
- Ⓓ $4x^3y^5$
- Ⓔ $682.7x^3y^5$

50. **Performance Task** A milk processing company uses cylindrical-shaped containers. The height of the container is equal to the diameter of the base.



Part A How much material is needed to make the lateral surface area of the shipping container?

Part B The cargo hold of a ship is 20 ft high. What is the largest number of these shipping containers that could be stacked on top of each other inside the cargo hold?